Wen Huang

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Research Interests

Speech and audio signal processing, domain adaptation and generalization, self-supervised learning

Education

Shanghai Jiao Tong University Shanghai, China
M.S. in Electronic and Information Engineering, GPA: 3.93/4.0 Sept. 2023 – Present
B.E. in Information Engineering, GPA: 3.85/4.0 Sept. 2019 – Jun. 2023

Relevant Coursework: Machine Learning, Deep Learning, Intelligent Speech Technology, Probability and Statistics, Linear Algebra, Mathematical Analysis, Graph Theory, Optimization, Numerical Analysis, Information theory and Coding, Digital Signal Processing, Data Structure and Algorithm, Programming Languages Theory, Computer Organization and Architecture, Computer Networks, Cloud Computing, Parallel Data Processing

Research Experience

Shanghai Jiao Tong University Shanghai, China
Research Assistant, advised by Prof. Yanmin Qian Jun. 2022 – Present

Speaker verification | *self-supervised learning, domain adaptation and generalization*

- Explored self-supervised contrastive learning methods for speaker verification tasks, enhancing system performance with limited labeled data to achieve parity with fully supervised systems.
- Proposed and developed an unsupervised domain adaptation framework using pseudo labeling for speaker verification, yielding consistent improvements across various domain adaptation scenarios.
- Designed a domain-specific adaptable module to mitigate domain variances and thus bolster the performance of speaker verification systems in multi-domain environments.

Audio classification and detection | *semi-supervised and self-supervised learning*

- Implemented large-scale pre-trained speech and audio models to enhance anomaly sound detection tasks, diverging from conventional reconstruction-based methods for improved detection accuracy.
- Investigated semi-supervised strategies and adaptation techniques for acoustic scene classification to address challenges associated with label scarcity and domain shift.

Publications

Robust Cross-Domain Speaker Verification with Multi-Level Domain Adapters.
W. Huang, B. Han, S. Wang, Z. Chen and Y. Qian. in ICASSP2024 [paper]

Exploring Large Scale Pre-Trained Models for Robust Machine Anomalous Sound Detection.
B. Han, Z. Lv, A. Jiang, **W. Huang**, Z. Chen, Y. Deng et al. in ICASSP2024 [paper]

Improving Dino-Based Self-Supervised Speaker Verification with Progressive Cluster-Aware Training
B. Han, **W. Huang**, Z. Chen, and Y. Qian. in ICASSPW2023 [paper]

Honors and Awards

Third Place, ICME2024 Acoustic Scene Classification Grand Challenge	2024
Outstanding Graduates in Shanghai Jiao Tong University	2023
Huatai Securities Science & Technology Scholarship	2022
SPEIT Academic Second-class Scholarship	2021
A Class Scholarship in Shanghai Jiao Tong University	2020

Professional Affiliations

Institute of Electrical and Electronics Engineers (IEEE)	Graduate Student Member
IEEE Signal Processing Society	Member
IEEE Women In Engineering	Member

Skills

Languages: C/C++, Python, Java, JavaScript, Matlab
Tools: Git/GitHub, Linux/Unix Shell, Docker, AWS
Frameworks: Pytorch, Tensorflow

Teaching Experience

Teaching Assistant, *English Speech and Techniques*, SJTU	2024
Teaching Assistant, *Practical English for Going Abroad* at SJTU	2024

Service

Volunteer at Shanghai Metro Stations	2023-2024
Head of Design Department, Volunteer Corps, SJTU	2020-2021
Volunteer at Shanghai K-11 Art Museum	2019-2020